

Disability, Chronic Conditions, And Labor Supply

Job Choice, Accommodation, And Dynamic Exit

Special Topic 6: Labor Market, Health, and Population – Week 2

Central question

- How do disability and chronic conditions reshape labor-force attachment, hours, occupation, accommodations, and exit?
- The labor objects are employment, hours, wages, occupation, job amenities, mobility, employer response, and welfare.
- Week 2 discipline: separate capacity, program rules, accommodation, local demand, and selection.

Why this is labor economics

- Disability changes the feasible set for work, not only clinical status.
- Chronic conditions can alter hours, search, commuting, schedules, task assignment, and promotion.
- Firms respond through accommodation, screening, retention, reassignment, and job design.
- Program rules and benefit receipt matter because they change the employment boundary.

Five mechanisms to keep separate

Mechanism	What moves	Typical outcome
Capacity	tasks, stamina, cognition, pain, commute	hours, exit
Programs	eligibility, benefits, reassessment risk	claiming, work
Accommodation	schedules, tasks, leave, remote work	retention
Local demand	feasible employers and occupations	mobility
Selection	who applies, receives, or is observed	bias

- A single employment coefficient can bundle all five mechanisms.

Direct capacity effects vs program incentives

$$V^W(H_{it}, A_{it}, D_{mt}) \geq V^B(H_{it}, R_{it}, B_{it})$$

- Capacity effects: health limits tasks, feasible hours, commuting, schedules, or productivity.
- Program effects: eligibility, earnings tests, waiting periods, benefits, and reassessment risks change the value of work.
- Disability status is not pure health; it blends impairment, rules, application costs, and local opportunities.
- Examiner, threshold, and reassessment designs identify program response for marginal cases.

Dynamic constraints

Object	Immediate margin	Dynamic margin
Acute onset	absence, hours cut	recovery or persistent exit
Chronic condition	task and schedule friction	career scarring, downgrading
Functional limitation	feasible jobs shrink	lower mobility, amenity demand
Benefit receipt	award and work incentives	reentry and attachment
Accommodation	retention today	match-specific dependence

- The question is always: which horizon and which labor margin?

Short-run vs long-run effects

- Short run: missed work, reduced hours, urgent medical spending, leave decisions, liquidity stress, and temporary accommodation.
- Medium run: job retention, employer reassignment, treatment response, benefit application, and search.
- Long run: persistent exit, occupational downgrading, lower income and consumption, weaker mobility, and welfare loss.
- A one-year employment estimate, five-year earnings estimate, and ten-year consumption estimate are different estimands.

Treatment, accommodation, return to work

Response	Labor-market channel	Main identification issue
Treatment	improves capacity or stabilizes symptoms	selection into care
Accommodation	changes feasible tasks and schedules	endogenous firm response
Return-to-work policy	lowers reentry risk or benefit loss	program-margin LATE
Remote/flexible work	relaxes commute and schedule constraints	occupation sorting

- These are not outside the labor model; they change the feasible set and the value of work.

Employer response and local demand

- Strong local demand can absorb health limitations through sorting, flexibility, and outside offers.
- Weak local demand can turn the same limitation into nonemployment or benefit application.
- Firms may retain and accommodate valuable workers, or screen and assign health-limited workers to lower-return tasks.
- Selection into observed disability status depends on severity, job loss, information, application costs, and expected benefits.

Monopsony, job lock, and worker inelasticity

Channel	Why outside options shrink	Labor implication
Insurance lock	leaving risks coverage and treatment continuity	lower quits
Accommodation lock	current setup may not transfer	employer-specific rents
Local scarcity	few feasible jobs or commutes	lower elasticity
Stigma/uncertainty	new employers may not accommodate	search costs
Information advantage	current firm knows productivity and limits	wage-setting power

- Lower mobility may reveal lock-in, not better matches.

Frontier methods

Design	What it can identify	Caution
Acute-shock event study	disruption and recovery after sharp health events	severity
Disability-onset admin event study	dynamic paths after observed onset	institutional status
Threshold/rule design	program incentives and receipt effects	local margin
Twin/sibling comparison	family endowment confounding	external validity
Genetic/MR design	chronic-condition channels	exclusion restriction
Matched employer-employee	firm accommodation and retention heterogeneity	sorting
Treatment reform	care, accommodation, or reentry response	bundled changes

- Primary anchor: Meyer and Mok on long-run disability incidence for earnings, income, and consumption.
- Challenge anchor: Hill, Maestas, and Mullen on employer accommodation and labor supply.
- Reproduce: estimate a synthetic disability-onset event-study profile.
- Diagnose: classify direct capacity, program, accommodation, local-demand, and selection channels.
- Transfer: redesign the logic for acute shocks, treatment access, program thresholds, or matched-firm accommodation.

Research design checklist

- ① Name the health object: onset, diagnosis, function, chronic burden, certification, or benefit receipt.
- ② Name the labor margin: employment, hours, earnings, occupation, mobility, accommodation, or welfare.
- ③ State the horizon: immediate disruption, medium-run adjustment, or long-run scarring.
- ④ Separate capacity from programs, firms, local demand, and selection.
- ⑤ Say what welfare object remains invisible in wages or employment.

- Week 2 centers working-age health limitations and the worker-employer match.
- Week 3 broadens the unit to household timing and population events.
- The next objects are fertility, mortality, caregiving, aging, and lifecycle labor allocation.

Takeaways

- Disability and chronic conditions change capacity, job choice, amenities, program interaction, and employer response.
- Short-run disruption and long-run scarring are different estimands.
- Accommodation and treatment can change labor supply; they are not merely background policy.
- Job lock and accommodation dependence can make labor supply more inelastic and strengthen employer power.